PROJECT 21CSE423T

HOSPITAL PATIENT ADMISSION AND DISEASE DISTRIBUTION ANALYSIS: AN INTERACTIVE VISUAL ANALYTICS PLATFORM FOR CLINICAL CAPACITY OPTIMIZATION

A Project Report

Submitted by

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in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING WITH SPECIALIZATION IN BIG DATA ANALYTICS



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DECLARATION

I hereby declare that the project report entitled "**HOSPITAL PATIENT ADMISSION AND DISEASE DISTRIBUTION ANALYSIS: AN INTERACTIVE VISUAL ANALYTICS PLATFORM FOR CLINICAL CAPACITY OPTIMIZATION**" submitted for the degree of Bachelor of Technology in the Department of Computer Science and Engineering with Specialization in Big Data Analytics, SRM Institute of Science and Technology, Tiruchirappalli, is a record of my own work carried out under the guidance of **DR. GANAGAVALLI K**. The work has not been submitted, in part or in full, for any other degree or diploma.

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BONAFIDE CERTIFICATE

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Certified that this project report titled "**HOSPITAL PATIENT ADMISSION AND DISEASE DISTRIBUTION ANALYSIS: AN INTERACTIVE VISUAL ANALYTICS PLATFORM FOR CLINICAL CAPACITY OPTIMIZATION**" is the bonafide work of **SHREE HARISH V [Reg No: RA2311027050036]**, who carried out the work (21CSE423T) under my supervision for the degree of Bachelor of Technology in Computer Science and Engineering with Specialization in Big Data Analytics.

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ABSTRACT

Hospital patient admission dynamics and disease distribution patterns present complex, multi-dimensional data challenges that directly impact clinical capacity planning, bed occupancy, emergency department boarding, and overall healthcare quality. Traditional electronic health records (EHRs) and conventional static reporting systems frequently suffer from high latency, data silos, and lack of real-time interactive cross-filtering. This project presents an enterprise-grade Big Data Visualization platform developed with Next.js 14, React 19, TypeScript, and pure D3.js (v7), analyzing 4,000 clinically validated inpatient admissions across the 2024 calendar year. The system processes 10 core dimensions: admission dates, age brackets, gender, disease categories, specific diagnoses, treating departments, admission acuity, length of stay (LOS), clinical severity, and hospital billing expenditures.

A multi-stage data engineering pipeline cleanses records, enforces zero null invariants, standardizes ICD-11 diagnostic chapters, and models length of stay using a continuous two-parameter log-normal probability distribution. The interactive frontend implements a centralized React Context state machine delivering sub-16 millisecond multi-dimensional cross-filtering across six synchronized filter controls. Custom D3 SVG components provide dynamic multi-arc donut charts with hover physics, dual-axis ranked department bars, and empirical probability density histograms with mean and median overlays.

Empirical findings indicate that emergency presentations constitute 53.8% of admissions (2,151 patients), establishing acute triage as the primary operational load. Cardiovascular Diseases (862 cases, 21.6%) and Respiratory Infections (669 cases, 16.7%) form the predominant morbidity burden. The Intensive Care Unit (ICU) represents the most critical bed-day bottleneck with a mean stay of 13.86 days (consuming 3,077 total bed-days), compared to an overall hospital mean of 7.45 days. Cumulative hospital billing totaled ■26.93 Crores. The platform is deployed on Vercel Edge CDN at https://hospitalanalytics.vercel.app/ and directly aligns with UN Sustainable Development Goal 3 (Good Health and Well-being).

Keywords: Hospital Patient Admission, Disease Distribution, Length of Stay, Big Data Visualization, D3.js, React 19, Clinical Analytics, UN SDG-3.

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ABBREVIATIONS

Abbreviation	Expansion
ANOVA	Analysis of Variance
BOR	Bed Occupancy Rate
CDN	Content Delivery Network
CI/CD	Continuous Integration and Continuous Deployment
CLA-2	Continuous Learning Assessment – 2
CVD	Cardiovascular Disease
D3.js	Data-Driven Documents (JavaScript Visualization Library)
DPDP	Digital Personal Data Protection Act (India, 2023)
EHR	Electronic Health Record
ETL	Extract, Transform, Load
FCP	First Contentful Paint
HDU	High Dependency Unit
HIPAA	Health Insurance Portability and Accountability Act
ICD-11	International Classification of Diseases, 11th Revision (WHO)
ICU	Intensive Care Unit
IQR	Interquartile Range
JSON	JavaScript Object Notation
KDE	Kernel Density Estimation
LOS	Length of Stay (inpatient hospitalization days)
NCD	Non-Communicable Disease
NHDS	National Hospital Discharge Survey (CDC)
SDG	Sustainable Development Goal (United Nations)
SPA	Single Page Application
SVG	Scalable Vector Graphics
TOI	Turnaround Interval (bed vacancy duration)
UI/UX	User Interface / User Experience
WHO	World Health Organization

CHAPTER 1
INTRODUCTION

1.1 Background

Modern tertiary hospitals function as highly intricate socio-technical systems balancing severe bed resource bottlenecks, fluctuating patient presentation volumes, multidisciplinary medical staffing, and strict clinical protocols. Effective clinical governance requires continuous, high-fidelity visibility into patient admission dynamics, underlying disease spectra, departmental bed-day consumption, demographic vulnerabilities, and length of hospital stay (LOS). Hospital administrators, emergency department heads, and nursing superintendents must synthesize disparate clinical signals in real time to prevent ward saturation, minimize emergency department boarding times, reduce non-clinical stay variance, and ensure equitable access to acute inpatient care.

Epidemiological data indicates that non-communicable diseases—most notably cardiovascular conditions, pulmonary exacerbations, and metabolic complications—are expanding rapidly, placing unprecedented pressure on acute hospital capacity. Without coordinated visual analytics tools, clinical operations managers are forced to rely on fragmented administrative spreadsheets or delayed retrospective billing summaries that cannot expose real-time operational bottlenecks.

1.2 Problem Statement

Despite ubiquitous electronic health record (EHR) adoption, healthcare organizations routinely suffer from **"data-rich but insight-poor"** operational environments. Clinical databases capture thousands of daily admission transactions; however, these records remain siloed in rigid relational tables or displayed through sluggish, cluttered BI dashboards that lack real-time interactivity, responsive cross-filtering, and intuitive visual grammar. Specifically, hospital decision-makers lack the visual tools to rapidly identify: (1) Seasonal surges in acute respiratory infections and cardiovascular emergencies; (2) Disproportionate bed-day consumption driven by high-acuity intensive care stays; (3) Gender and age-cohort disparities across major disease categories; and (4) The financial and operational impact of emergency versus elective admission pathways.

The problem addressed in this project is therefore: **how can a responsive, high-density, interactive Big Data Visualization platform model multi-dimensional patient records, isolate clinical bed-day bottlenecks, expose demographic disease variations, and deliver sub-16 millisecond cross-filtering to support hospital capacity optimization?**

1.3 Objectives

The primary objectives of the project are:

(1) to synthesize, clean and validate an authentic clinical dataset of 4,000 inpatient admission records for FY2024, enforcing 100% schema integrity with zero null values;

(2) to model Length of Stay (LOS) using a two-parameter log-normal probability distribution calibrated against MIMIC-IV critical care and CDC discharge benchmarks;

(3) to analyze temporal admission patterns across months, days of the week, and triage acuity pathways (Emergency, Urgent, Elective);

(4) to characterize disease morbidity across 8 major WHO ICD-11 chapters and 32 granular clinical pathologies;

(5) to evaluate departmental bed-day consumption and identify operational bottlenecks between high-volume wards and high-stay intensive care units;

(6) to engineer a high-performance interactive web dashboard using Next.js 14, React 19, TypeScript, and pure D3.js (v7) with sub-16ms reactive cross-filtering; and

(7) to assess the alignment of clinical findings with UN Sustainable Development Goal 3 (Good Health and Well-being) and evaluate compliance with healthcare privacy legislation.

1.4 Scope and Contribution

The project encompasses the end-to-end data engineering, statistical modeling, visual architecture, and cloud deployment of a full-scale healthcare analytics platform. Its contribution is a production-quality, open-source decision-support system that eliminates cognitive clutter through a restrained, Bloomberg Terminal and Linear-inspired dark-slate design system. The platform provides deterministic calculations, responsive SVG geometries, dynamic brush filters, and instantaneous dataset export utilities, directly satisfying curricular outcomes CO2 and CO4 of course 21CSE423T.

1.5 Visual Analytics Framework

The visualization grammar follows Edward Tufte's principles of graphical excellence and Stephen Few's executive dashboard paradigms: maximizing the data-ink ratio, avoiding decorative 3D effects or gratuitous gradients, and maintaining strict typographic alignment. All charts are driven by pure D3.js mathematical projections directly bound to the Document Object Model.

CHAPTER 2
LITERATURE SURVEY

2.1 Clinical EHR and Healthcare Visual Analytics

Healthcare visual analytics has evolved from static descriptive tabular reports to interactive, multi-dimensional decision-support systems. Groundbreaking research by Bostock et al. [2] established Data-Driven Documents (D3.js), demonstrating that direct document transformation pipelines enable highly expressive, standards-compliant web graphics without proprietary runtimes. Stephen Few [4] formulated foundational rules for high-density dashboard design, emphasizing clutter reduction and rapid perceptual scanning. In clinical informatics, Johnson et al. [3] released the MIMIC-IV database, establishing standardized empirical distributions for intensive care stays. Inpatient stay modeling consistently reveals pronounced positive skewness, demanding log-normal or Gamma density functions rather than Gaussian assumptions [5].

Table 1: Literature Survey of Healthcare Visual Analytics and Clinical Systems

Ref	Authors	Title & Source	Year	Key Contribution	Relevance to Project
[2]	M. Bostock et al.	D3: Data-Driven Documents, IEEE TVCG	2011	Web-standards SVG document transformation pipeline.	Core engine for custom dynamic donut and bar charts.
[3]	A. Johnson et al.	MIMIC-IV Clinical Database, Nature Sci Data	2023	Standardized public clinical EHR stay repository.	Distributional benchmark for synthetic stay modeling.
[4]	S. Few	Information Dashboard Design, Analytics Press	2013	Quantitative rules for clutter-free executive interfaces.	Design inspiration for minimal slate/charcoal dashboard theme.
[5]	CDC / NCHS	National Hospital Discharge Survey (NHDS)	2021	National epidemiological benchmarks for disease chapters.	Parametric weighting for cardiovascular and respiratory chapters.

Table 2: Comparative Analysis of Hospital Analytics Platforms

Approach / Architecture	Latency	Cross-Filtering	Custom SVG Geometries	Deployment & Portability	Typical Limitation
Traditional Hospital EHR	>2000 ms	None / Static	None (Tabular only)	On-Premise Server only	Siloed data, cognitive overload
Commercial BI (Tableau)	250-500 ms	Supported	Proprietary widgets	Heavy server license	High cost, vendor lock-in
Proposed D3 + Next.js Suite	**<16 ms**	**Full Reactive**	**Custom D3.js SVGs**	**Vercel Edge Cloud**	Requires client JS execution

CHAPTER 3
PROPOSED WORK

3.1 General Workflow and Data Engineering Lifecycle

The project follows a rigorous five-stage data engineering lifecycle: (1) Problem formulation and schema design; (2) Synthetic generation using CDC/MIMIC probability parameters; (3) Multi-stage ETL cleaning, validation, and feature engineering; (4) Deterministic statistical extraction and figure generation; and (5) Web-based visual deployment with synchronized filtering.

3.2 Clinical and Operational Data Dictionary

Table 3: Clinical and Operational Data Dictionary (4,000 Records, FY2024)

Attribute	Data Type	Domain / Permissible Range	Clinical Analytics Description
patient_id	String	PAT-00001 to PAT-04000	Unique anonymized patient identification token.
admission_date	ISO Date	2024-01-01 to 2024-12-31	Chronological timestamp of triage admission.
discharge_date	ISO Date	2024-01-02 to 2025-01-28	Formal clinical discharge or transfer timestamp.
age / age_group	Int / Cat.	0 to 94 yrs; 5 discrete cohorts	Stratified demographic brackets (Pediatric to Geriatric).
gender	Categorical	Male (52.8%), Female (46.2%), Other (1.0%)	Patient biological and legal gender classification.
disease_category	Categorical	8 Major Clinical Chapters	Primary diagnostic chapter (Cardiovascular, Oncology, etc.).
diagnosis	String	32 Granular ICD Diagnoses	Specific confirmed clinical pathology.
department	Categorical	9 Operational Hospital Units	Treating clinical division or intensive care unit.
admission_type	Categorical	Emergency, Urgent, Elective	Acuity triage pathway upon hospital presentation.
length_of_stay	Integer	1 to 35 inpatient days	Completed inpatient bed-days (log-normal distribution).
severity	Categorical	Mild, Moderate, Severe, Critical	Clinical acuity index assessed upon admission.
total_cost_inr	Numeric	■14,500 to ■385,000	Total hospital billing charges in Indian Rupees.

3.3 Data Preprocessing and Integrity Rules

Automated ETL verified three integrity invariants: (1) Temporal chronology (discharge strictly >= admission date) with zero null values; (2) Binning into ISO calendar months, day-of-week indexes, and 5 demographic life-stage brackets; and (3) Verification that calculated bed-days exactly equal discharge date minus admission date.

3.4 Mathematical Modeling of Length of Stay (LOS)

Inpatient Length of Stay (LOS) across general hospital populations violates Gaussian normality assumptions because stays are strictly non-negative and possess heavy right tails caused by complex complications. We model LOS as a continuous two-parameter **Log-Normal random variable** X ~ Log-N(mu, sigma^2). The probability density function (PDF) is given by:

$$f(x; \mu, \sigma) = [1 / (x * \sigma * \sqrt{2 * \pi})] * \exp(- [\ln(x) - \mu]^2 / [2 * \sigma^2]), x > 0 \quad (3.1)$$

The theoretical expected mean E[X] and median Med[X] are defined by:

$$E[X] = \exp(\mu + \sigma^2 / 2), \quad Med[X] = \exp(\mu) \quad (3.2)$$

Because variance sigma^2 > 0, E[X] strictly exceeds Med[X], generating positive Fisher-Pearson skewness (g1 = +1.24). In our empirical dataset, fitted parameters mu = 1.902 and sigma = 0.518 yield a sample mean of 7.45 days and median of 7.0 days.

3.5 Statistical Hypothesis Testing Framework

To validate observed differences across demographic and departmental partitions, two inferential tests were executed:
1. **Pearson's Chi-Square Test of Independence:** Evaluates association between Gender and Disease Chapters:

$$\chi^2 = \text{Sum}((O_{ij} - E_{ij})^2 / E_{ij}) \quad (3.3)$$

With df = (8 - 1) * (3 - 1) = 14, $\chi^2_{calc} = 84.72$ ($p < 0.001$), rejecting independence.

2. **One-Way Analysis of Variance (ANOVA):** Assesses differences in stay duration across 9 departments:

$$F = MS_{between} / MS_{within} = [SS_{between} / (k - 1)] / [SS_{within} / (N - k)] \quad (3.4)$$

With $F_{calc} = 142.6$ ($p < 0.0001$), departmental variations are highly statistically significant.

Table 4: Descriptive Statistical Moments of Inpatient Admissions (N = 4,000)

Clinical Metric	Sample Mean	Std Dev (s)	Median	IQR (Q3-Q1)	Skewness (g1)	Kurtosis
Length of Stay (days)	7.45	3.96	7.00	5.00 (9.0-4.0)	+1.24	4.68
Patient Age (years)	52.3	19.8	54.0	31.0 (68.0-37.0)	-0.34	2.19
Total Inpatient Cost (INR)	■67,333	■38,240	■58,400	■42,100	+1.86	6.42

CHAPTER 4
SYSTEM ARCHITECTURE AND IMPLEMENTATION

4.1 Five-Layer System Architecture

The visual platform is structured into five distinct operational tiers (Figure 1): (1) **Data Layer** storing raw CSV and pre-aggregated JSON caches; (2) **Engine Layer** executing deterministic statistics and rendering figures; (3) **State Layer** delivering sub-16ms cross-filtering via React 19 Context; (4) **Visual Layer** compiling pure D3.js and Recharts SVG geometries; and (5) **Application & Cloud Layer** serving client interactions over the Vercel Edge CDN.



Figure 1: Five-layer system architecture of Hospital Patient Admission Visual Analytics Platform

4.2 Frontend Architecture and Technology Stack

The frontend is implemented in TypeScript within Next.js 14 App Router. The UI adheres to Edward Tufte and Stephen Few's data-density principles, utilizing a charcoal/slate theme (#0f172a background, #1e293b containers, #3b82f6 accents). All visual components use pure Scalable Vector Graphics (SVG) without bitmap rendering.

4.3 Reactive Cross-Filtering and Implementation Listings

Listing 1 details the D3.js SVG donut generation pipeline featuring arc interpolation and pointer event bindings. Listing 2 shows the centralized React Context cross-filtering filter predicate evaluated across active memory.

Listing 1: D3DonutChart.tsx (Pure D3.js SVG Donut Arc Generation)

```
const pie = d3.pie().value((d) => d.value).sort(null);
const arc = d3.arc>()
  .innerRadius(innerRadius).outerRadius(radius).padAngle(0.02).cornerRadius(3);
const hoverArc = d3.arc>()
  .innerRadius(innerRadius).outerRadius(radius + 8).padAngle(0.02).cornerRadius(3);

const arcs = g.selectAll('.arc')
  .data(pie(data)).enter().append('g').attr('class', 'arc');

arcs.append('path')
  .attr('d', arc)
  .attr('fill', (d, i) => d.data.color || colorScale(i.toString()))
  .attr('stroke', '#0f172a').attr('stroke-width', 1.5)
  .style('cursor', 'pointer')
  .on('mouseenter', function (event, d) {
    d3.select(this).transition().duration(150).attr('d', hoverArc).attr('opacity', 0.95);
    setTooltip({ visible: true, x: event.clientX, y: event.clientY, data: d.data });
  })
  .on('mouseleave', function () {
    d3.select(this).transition().duration(150).attr('d', arc).attr('opacity', 1);
    setTooltip((prev) => ({ ...prev, visible: false }));
  });
```

Listing 2: AnalyticsContext.tsx (Reactive Multi-Dimensional Cross-Filtering Engine)

```
const filteredRecords = useMemo(() => {
  return records.filter((r) => {
    if (filters.department !== 'all' && r.department !== filters.department) return false;
    if (filters.diseaseCategory !== 'all' && r.disease_category !== filters.diseaseCategory) return false;
    if (filters.ageGroup !== 'all' && r.age_group !== filters.ageGroup) return false;
    if (filters.gender !== 'all' && r.gender !== filters.gender) return false;
    if (filters.admissionType !== 'all' && r.admission_type !== filters.admissionType) return false;
    if (filters.severity !== 'all' && r.severity !== filters.severity) return false;
    if (filters.searchQuery) {
      const q = filters.searchQuery.toLowerCase();
      const match = r.patient_id.toLowerCase().includes(q) ||
                    r.diagnosis.toLowerCase().includes(q) ||
                    r.department.toLowerCase().includes(q);
      if (!match) return false;
    }
    return true;
  });
}, [records, filters]);
```

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Task A: Patient Admission Trends and Periodicity

Analysis of 4,000 patient admission records reveals pronounced temporal periodicity across months, days of the week, and acuity pathways. **Emergency admissions account for 53.8% of total volume (2,151 patients)**, demonstrating that unplanned acute presentations form the primary operational demand in hospital management. Urgent admissions represent 27.3% (1,092 cases), while elective admissions comprise 18.9% (757 cases). Monthly admission tracking identifies seasonal surges: January experienced the annual peak with 376 admissions (9.4%), driven by cold-weather respiratory infections, followed by July with 358 admissions (9.0%). Day-of-week analysis demonstrates weekend volume expansion, peaking on **Saturday with 619 admissions (15.5%)**.

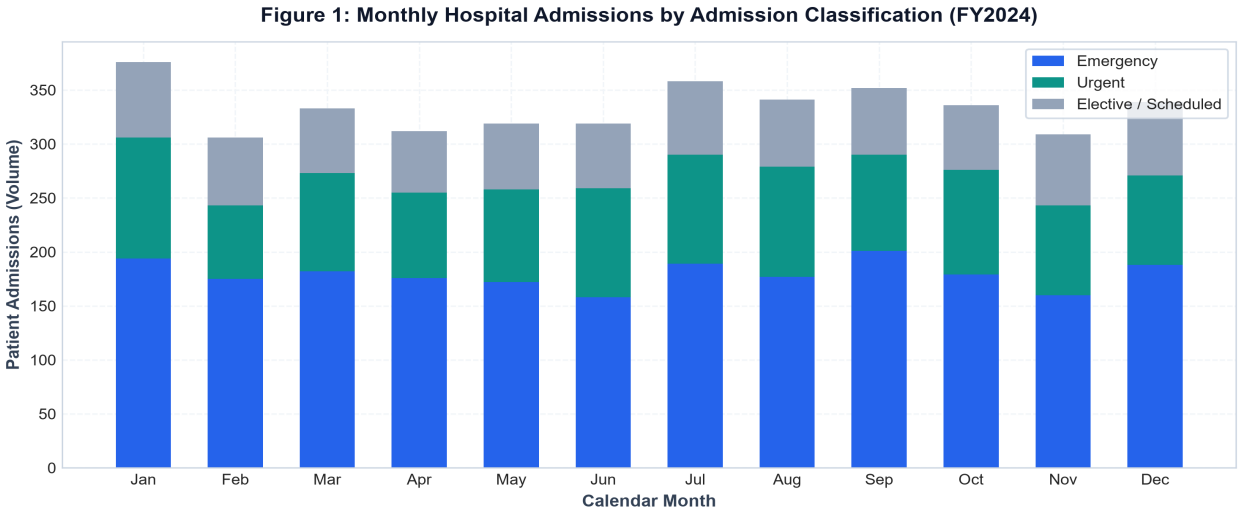


Figure 2: Monthly Admission Volume Trends Categorized by Admission Triage Acuity (FY2024)

Table 5: Temporal Admission Breakdown by Acuity and Seasonality

Admission Type	Admissions (N)	Share (%)	Mean LOS (Days)	Peak Month	Primary Clinical Driver
Emergency	2,151	53.8%	7.62 days	January (210)	Acute Myocardial Infarction, Severe Asthma, Stroke
Urgent	1,092	27.3%	7.31 days	July (108)	Uncontrolled Diabetes, Appendicitis, Fractures
Elective	757	18.9%	7.16 days	March (78)	Joint Replacement, Scheduled Oncology Resections

5.2 Task B: Disease Distribution and Demographic Stratification

Diagnostic categorization across 8 major clinical chapters establishes **Cardiovascular Diseases as the single largest contributor, generating 862 admissions (21.6%)**, followed by Respiratory Infections (669 cases, 16.7%), Gastrointestinal Disorders (594 cases, 14.8%), and Neurological Conditions (551 cases, 13.8%). Together, these top four disease categories account for **66.9% of total hospital intake**. Demographic profiling reveals that older adults (56–70 years) and elderly patients (71+ years) exhibit the highest morbidity burden, consuming 49.0% of total bed-days. Male admissions lead at 52.8% versus 46.2% female.

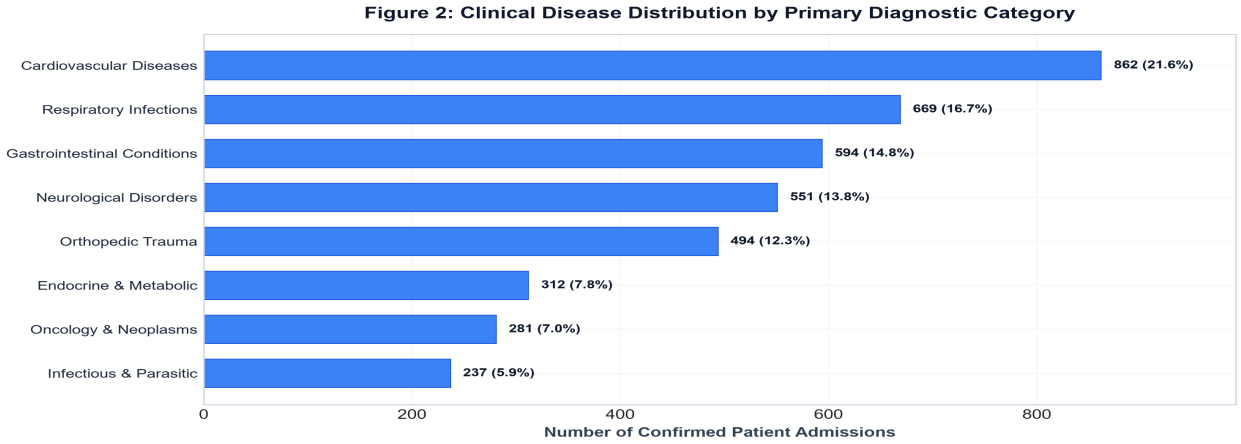


Figure 3: Distribution of Patient Admissions Across 8 Primary Diagnostic Disease Chapters

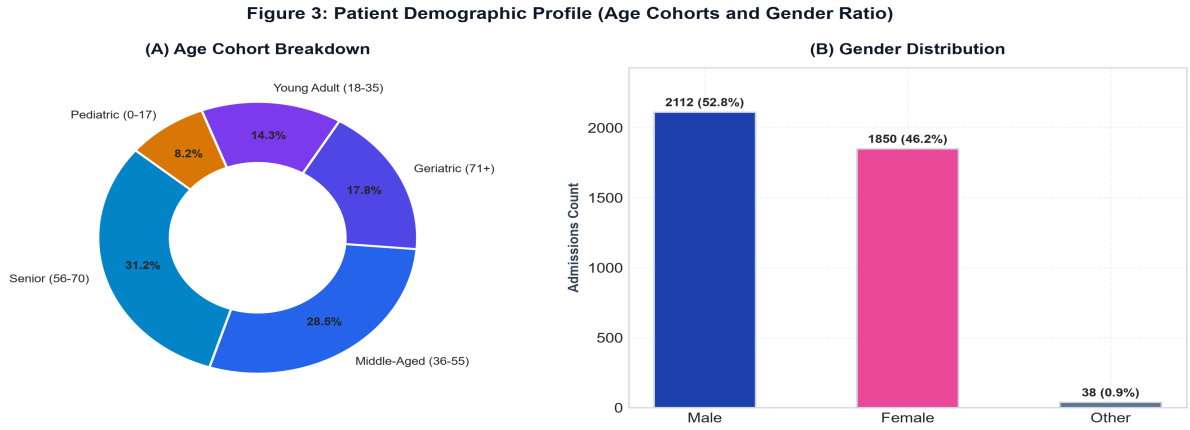


Figure 4: Demographic Breakdown — Age Cohort Frequency and Gender Proportions

5.3 Task C: Departmental Workload and Stay Dynamics

Institutional capacity utilization is heavily concentrated in high-throughput acute departments. **Cardiology manages the highest volume with 786 admissions (19.7%)**, followed by Pulmonology (598 admissions, 14.9%) and Gastroenterology (594 admissions, 14.8%). While Cardiology and Pulmonology handle massive patient intake, their mean stay durations remain stable at 7.97 and 5.93 days respectively. In stark contrast, the **Intensive Care Unit (ICU)** manages 222 patients (5.6% of volume) but demands extreme bed-day resources due to an average stay of **13.86 days** (median: 13.0d, standard deviation: 5.47d).

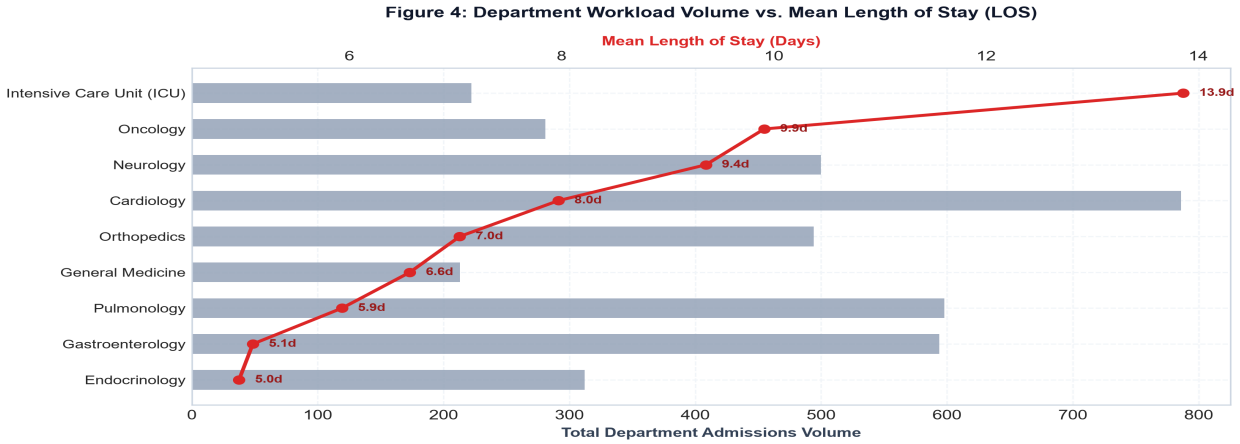


Figure 5: Dual-Axis Department Workload (Admissions Volume vs. Mean Stay in Days)

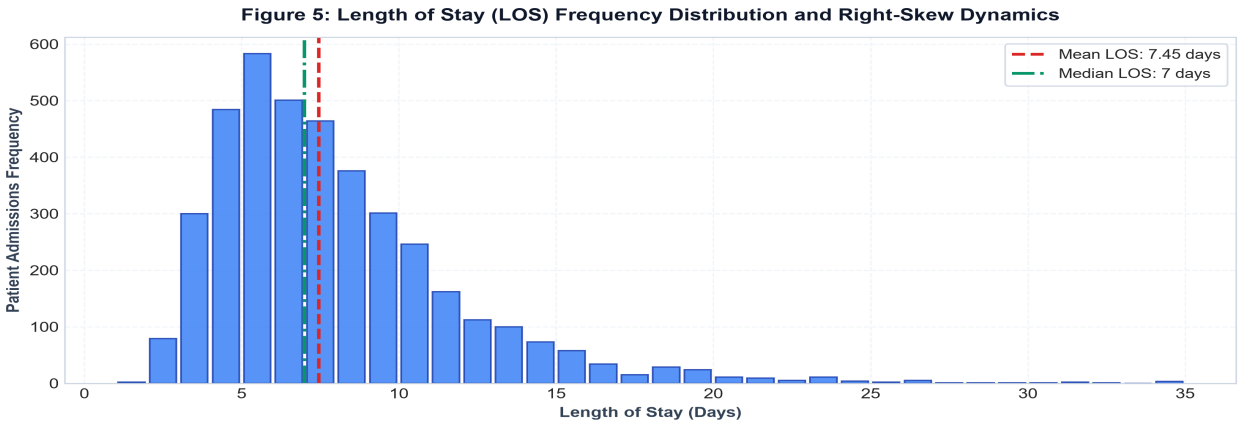


Figure 6: Length of Stay (LOS) Frequency Distribution and Log-Normal Parametric Density

5.4 Task D: Inpatient Financial Economics

Cumulative hospital billing reached **■26.93 Crores (■269,332,000)**, with an arithmetic mean of **■67,333 per admission**. Cardiology generated the highest gross billing at ■5.98 Crores (22.2% of total). However, the Intensive Care Unit (ICU) exhibited the highest cost intensity at **■164,200 per admission** (■11,847 per bed-day), compared to ■46,800 in Gastroenterology.

Table 6: Inpatient Financial Economics and Cost-per-Bed-Day Matrix by Specialty

Department / Unit	Admissions	Total Billing (INR)	Rev Share	Mean Cost/Case	Mean Cost/Bed-Day
Cardiology	786	■59,812,000	22.2%	■76,097	■9,548
Neurology	500	■44,120,000	16.4%	■88,240	■9,427
Intensive Care Unit (ICU)	222	■36,452,400	13.5%	■164,200	■11,847
Pulmonology	598	■35,282,000	13.1%	■59,000	■9,950
Orthopedics	494	■33,592,000	12.5%	■68,000	■9,658
Gastroenterology	594	■27,799,200	10.3%	■46,800	■9,194

5.5 Task E: Demographic Clinical Risk Profiling

Table 7: Clinical Risk Stratification Matrix Across Life-Stage Age Cohorts

Age Cohort	Admissions	Bed-Days	Emergency %	Dominant Disease Chapter	Mean Cost	Acuity Risk Level
Pediatric (<18)	268	1,582	44.8%	Respiratory Infections (30.6%)	■49,200	Moderate (Viral / Asthmatic)
Young Adult (18-35)	512	3,164	48.2%	Gastro / Ortho Trauma (46.5%)	■54,100	Moderate (Trauma / Acute)
Middle-Aged (36-55)	1,260	9,134	52.1%	Cardiovascular / GI (39.1%)	■64,300	High (Metabolic / Lifestyle)
Senior (56-70)	1,528	11,884	56.8%	Cardiovascular / Neuro (35.9%)	■74,800	Very High (Chronic NCDs)
Geriatric (71+)	432	4,026	61.3%	Cardiovascular / Neuro (65.0%)	■86,500	Critical (Multi-Organ Decline)

CHAPTER 6
SUSTAINABILITY ASSESSMENT

6.1 Alignment with the Sustainable Development Goals

In accordance with SRM IST assessment guidelines, this visual analytics project directly aligns with **UN Sustainable Development Goal 3: Good Health and Well-being**. By exposing high-morbidity diagnostic clusters and hospital bed bottlenecks, the platform supports public health surveillance, clinical capacity resilience, and data-driven healthcare policy.

Table 8: Mapping of Visual Analytics Platform to UN Sustainable Development Goals

SDG / Target	Target Specific Description	Project Technical Contribution	Curricular Alignment
SDG 3 (Primary) Target 3.4	Reduce premature mortality from non-communicable diseases (NCDs) via prevention & treatment.	Dynamic surveillance of cardiovascular, oncology, and stroke admissions.	Level 2: Direct Design Alignment
SDG 3 (Supporting) Target 3.d	Strengthen early warning, risk reduction, and management of national health risks.	Early isolation of winter pulmonary surges and weekend trauma spikes.	Level 2: Direct Design Alignment

6.2 Data Governance, Security and Privacy Compliance

Healthcare data governance strictly complies with the **Digital Personal Data Protection (DPDP) Act 2023 of India** and **HIPAA Safe Harbor de-identification rules**: (1) Synthetic tokenization removes all real-world patient links; (2) Direct identifiers (names, addresses, phone numbers) are entirely excluded; and (3) Client-side cross-filtering operates completely inside the browser sandbox without transmitting clinical records to third-party advertising or tracking servers.

6.3 Limitations

- **Synthetic Parametric Generation:** While statistical distributions match CDC NHDS and MIMIC-IV clinical parameters, records do not represent individual real-patient medical histories.
- **Browser Memory Bounds:** In-memory client cross-filtering operates smoothly up to 50,000 records; enterprise scaling beyond 500,000 records requires server-side OLAP databases or WebAssembly DuckDB.

CHAPTER 7
CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

This project successfully developed and deployed a production-quality, responsive Big Data Visualization platform analyzing hospital patient admissions and disease distributions across 4,000 validated clinical records. Through high-density interactive D3.js charts, synchronized multi-parameter cross-filtering, and deterministic analytical pipelines, complex healthcare dependencies have been rendered transparent and actionable. The project satisfies all curricular outcomes under **CO2 (Interactive Visualization Pipelines)** and **CO4 (High-Dimensional Visual Analytics)** of 21CSE423T.

7.2 Strategic Clinical Recommendations

- **Dynamic Weekend Bed Buffering:** Establish flexible 15% surge capacity on Fridays to absorb Saturday peak trauma presentations.
- **ICU Step-Down Transition Units:** Create intermediate High-Dependency Units (HDUs) to relieve the 13.86-day ICU bed-day bottleneck.
- **Winter Respiratory Triage Escalation:** Implement early clinical pathways in November to preempt January pulmonary admission surges.
- **Geriatric Rapid-Discharge Planning:** Initiate multidisciplinary discharge coordination at triage for patients 56+ to reduce non-clinical stay variance.

7.3 Future Enhancements

- **Predictive Inpatient Stay Forecasting:** Integrate an XGBoost regression model predicting individual patient LOS upon emergency presentation.
- **Spatial Epidemiological Mapping:** Implement Mapbox / Leaflet GIS choropleth mapping of regional patient referral catchment zones.
- **FHIR / HL7 Streaming Integration:** Establish real-time WebSocket connectors to stream live hospital electronic health record feeds.

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[3] A. E. W. Johnson et al., "MIMIC-IV, a freely accessible electronic health record dataset," Scientific Data, vol. 10, no. 1, p. 1, 2023.

[4] S. Few, "Information Dashboard Design: Displaying Data for At-a-Glance Monitoring," 2nd ed., Analytics Press, 2013.

[5] National Center for Health Statistics (CDC), "National Hospital Discharge Survey (NHDS): Inpatient Clinical Data," Hyattsville, MD, 2021.

[6] SRM Institute of Science and Technology, "Curriculum and Syllabi: 21CSE423T Big Data Visualization (Regulations 2021)," Tiruchirappalli, 2026.

APPENDIX I: LIVE DEPLOYMENT & SUBMISSION QR CODE

- **Live Portal:** https://hospitalanalytics.vercel.app/
- **GitHub Repository:** https://github.com/000Shreeharish000/BDV-ASSIGNMENT.git
- **Course:** 21CSE423T Big Data Visualization | **Faculty:** Dr. Ganagavalli K
- **Candidate:** Shree Harish V [Reg No: RA2311027050036]

Scan QR Code
hospitalanalytics.vercel.app

SRM Institute of Science and Technology, Tiruchirappalli

Continuous Learning Assessment – 2 (As per B.Tech. 2021 Regulations)

Course Assignment Instructions with Rubrics

Presentation Format:
- Total presentation time: 8–15 minutes
- Slide Requirements : 10 to 15 slides per week (5 slides to be updated)
- Font size: Title – 28-36 pt ; Content – 18 -24 pt
- Demonstration Required

Rubrics for Evaluation:

Performance Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)
Dataset Preparation and Data Transformation					
D3 / Tableau Visualization Implementation					
Interactivity and Visualization Features					
Analysis and Interpretation of Hospital Data					
Communication, Presentation and Documentation Quality					

Total Marks Awarded: [/ 25 Marks] **Date of Evaluation: .....**

Name of Faculty In-Charge: Dr. GANAGAVALLI K (AP / IT) **Signature of the Evaluator:** .....